

Lec 21 : Network layer numericals

1 | 11 | 23

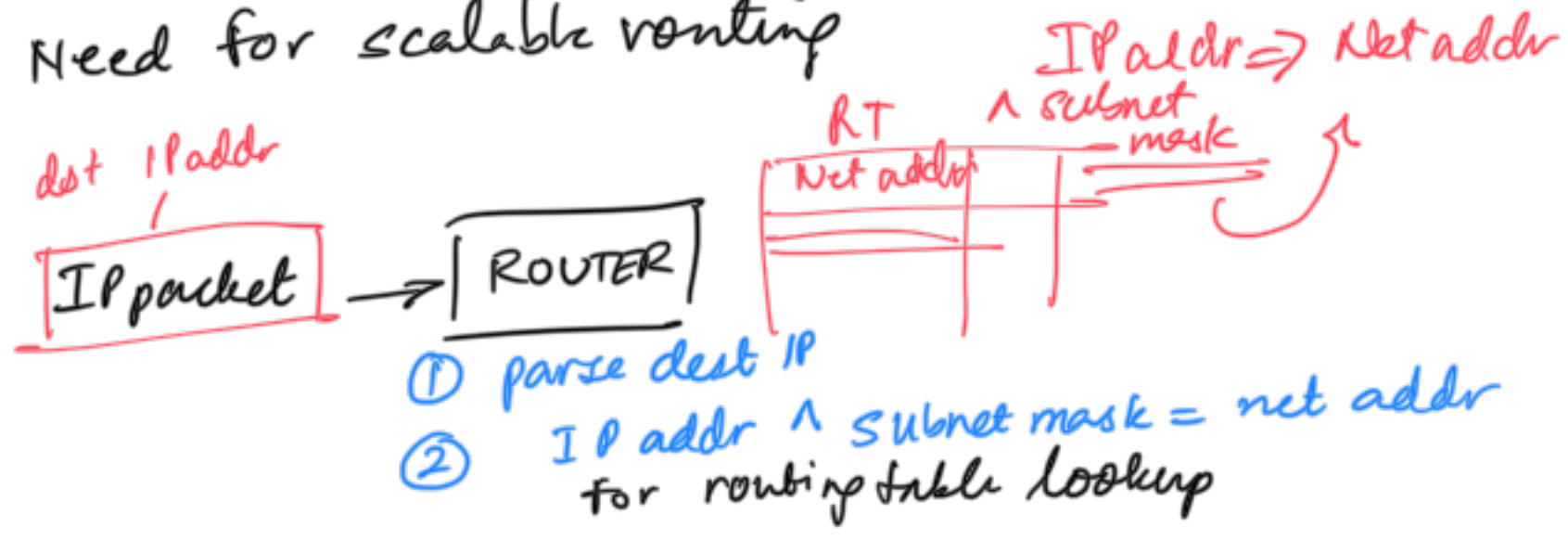
	Net bits	Host bits	Range
Class A	8	24	0.0.0.0 to 127.255.255.255
Class B	16	16	128.0.0.0 to 191.255.255.255
Class C	24	8	192.0.0.0 to 223.255.255.255

	Net bits	Host bits
Network addr	Specified (<u>AS IT IS</u>)	0...0 ←
Subnet mask	<u>1...1</u>	<u>0...0</u>
Broadcast addr	Specified or 1...1	1...1

Max. # hosts on a network = $2^n - 2$
 n : # of host id bits

Max # networks = 2^k
 k : # of network bits

* Need for scalable routing



Problem 1

nets = 30

hosts/net = 2000

(a)

Max # hosts/network : Class B : Hostid bits = 16 (m)

$$= 2^n - 2$$

$$= 65534$$

• Address utilization efficiency = $\frac{2000}{65535} = 3\%$

(b)

190.32.0.0 | 16

#netid k = 14

#hostid n = 16

networks = 30

#netid bits = $\log_2 30 = 5$ 32 bits



- ① 190.32.0.0
- ② 190.32.8.0
- ③ 190.32.16.0
- ⋮

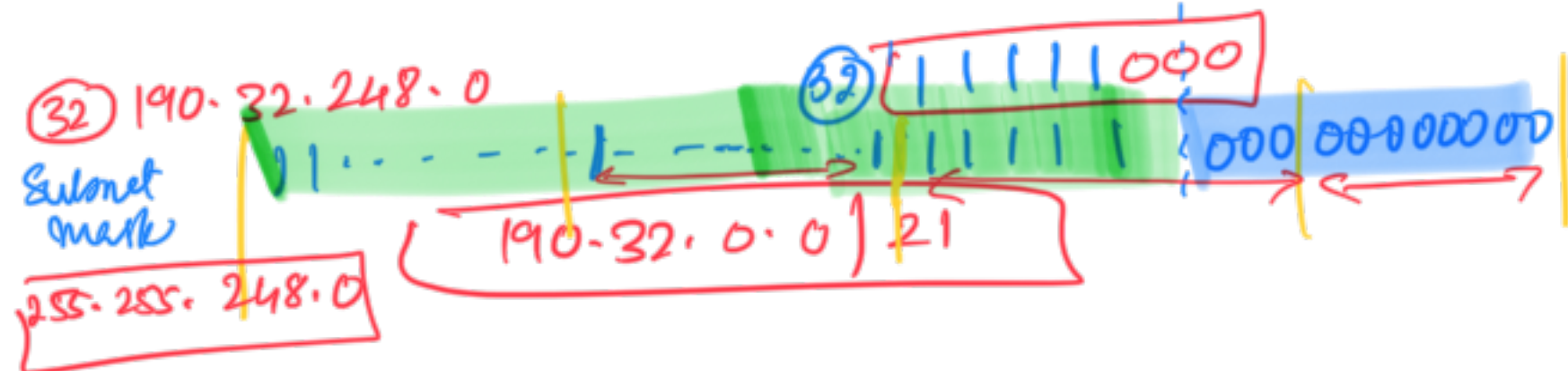
- ① 0000.0000
- ② 00001000
- 00010000
- ⋮
- 00011111
- ⋮
- 00100000
- ⋮
- ⋮
- ⋮

new n = 11

hosts/subnet

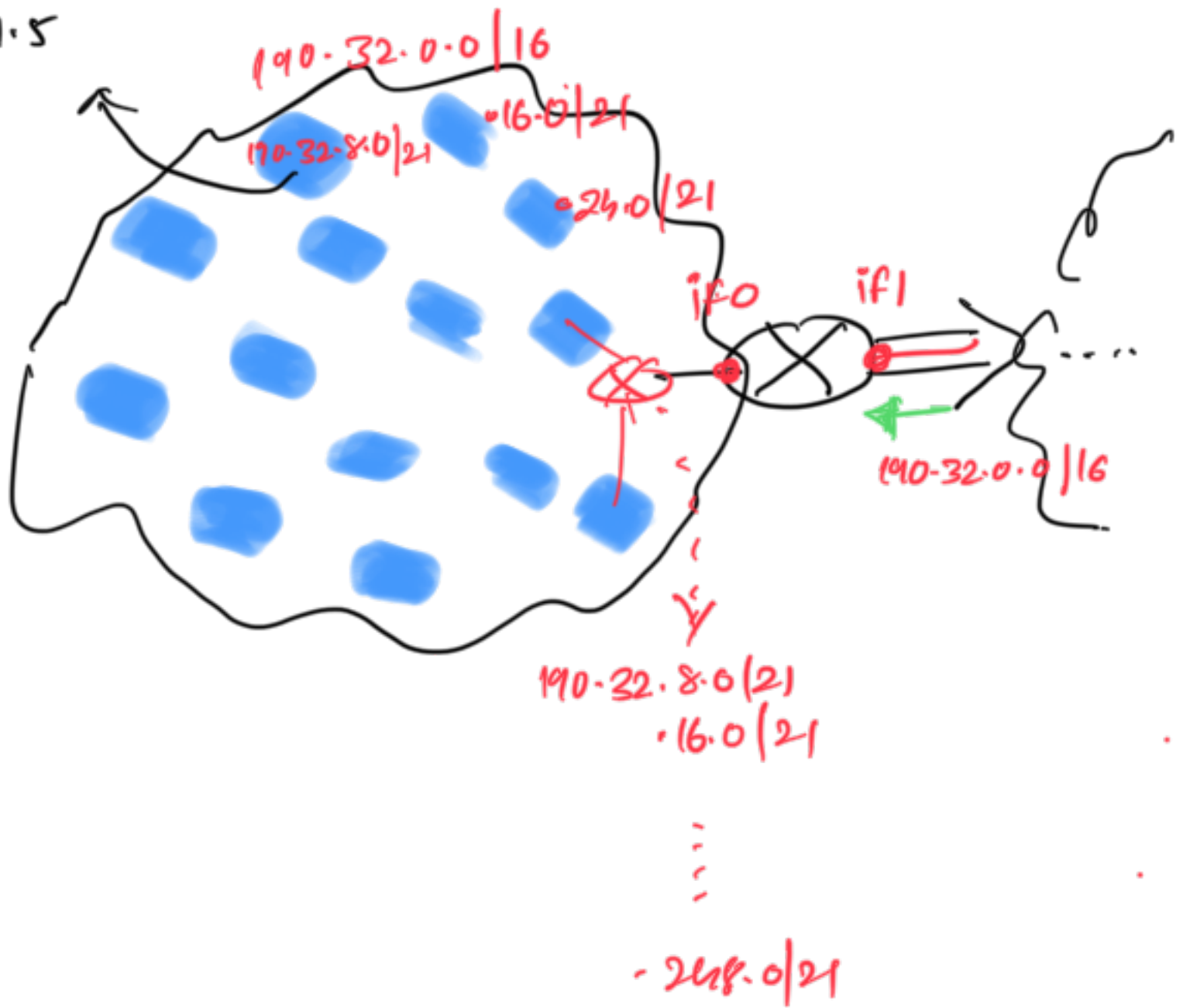
$$= 2^n - 2$$

$$= 2046$$



Addr util Efficiency = $\frac{2000}{2046}$

190.32.11.5



c) 190.32.11.5

1 255.255.248.0

dst IPaddr

subnet mask
(21)

11111000 248
00001011 11

00001000 8

190.32.8.0

Problem 2

hosts = 4000

If Class B addr is assigned

a)

$$\text{Efficiency} = \frac{4000}{65535} \approx 6.1\%$$

b) 227.7.16.0/24 Class C network

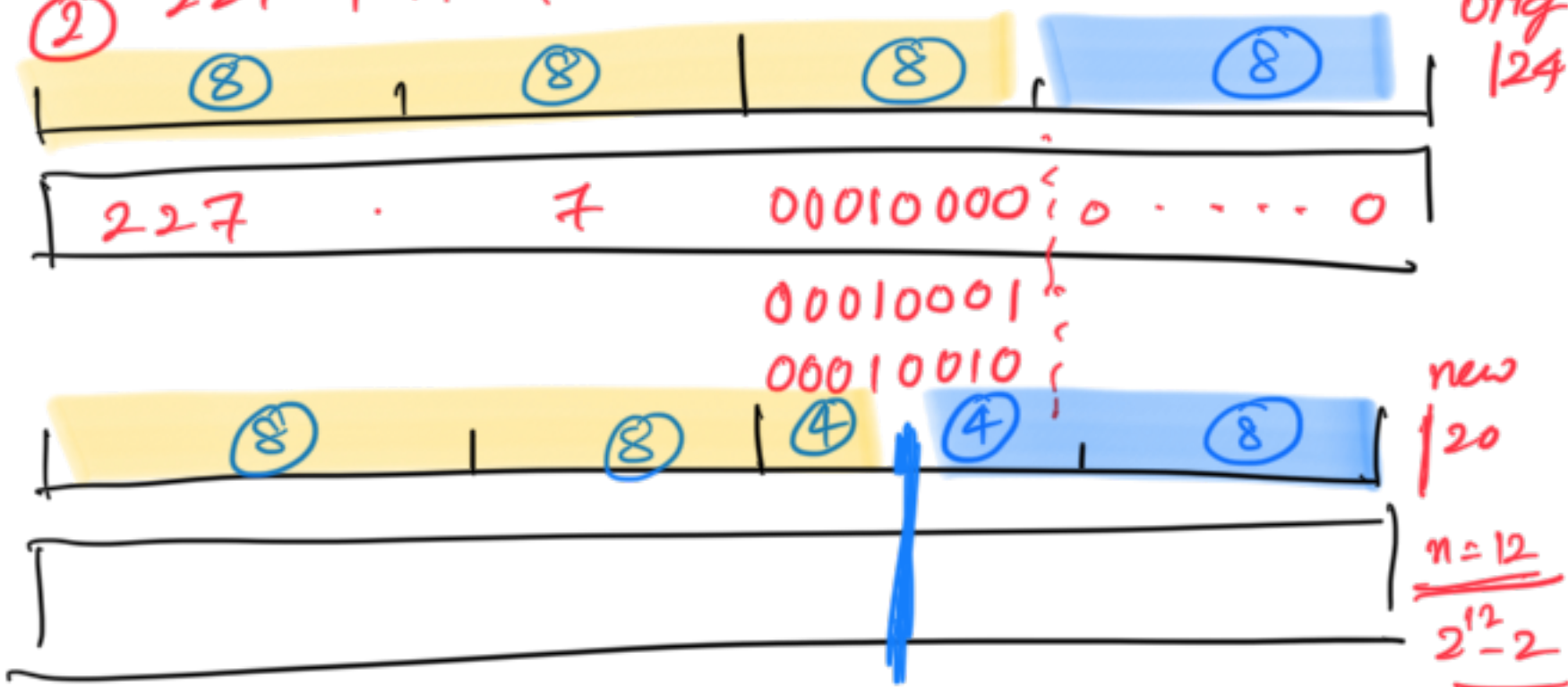
$$\# \text{ hostid, } n = 8 \quad \# \text{ hosts/net} = 2^8 - 2 = 254$$

$$\# \text{ Class C nets for 4000 hosts} = \frac{4000}{254} = \boxed{16}$$

① 227.7.16.0/24

..... ① 227.7.31.0/24

② 227.7.17.0/24



Problem 3

10.129.0.0 / 16

... Given

4 labs with # hosts

Lab A

200

Lab B

100

Lab C

50

Lab D

50

$$\# \text{ hosts} = 200 + 100 + 50 + 50 = 400$$

$$n = \log_2 400$$

$$n = 9$$

10 ✓		129 ✓	SUBNET		HOST	
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Lab A
200 hosts

10 129
10.129.0.0 / 24

00000000 0 00000000
00000000 0 11111111

Lab B
100 hosts

10 129
10.129.1.0 / 25

00000001 0 00000000
00000000 0 11111111

Lab C
50 hosts

10 129
10.129.1.128 / 26

00000001 10 00000000
00000000 10 11111111

Lab D
50 hosts

10 129
10.129.1.192 / 26

00000001 11 00000000
00000000 11 11111111

Problem 4:

Homework